

2024



Nationwide Scales Amazon EKS to Simplify and Accelerate IT Development

Nationwide Building Society ([Nationwide](#)) successfully transformed its IT landscape by implementing a centralized enterprise platform-as-a-service environment on AWS. With over 16 million members, the UK-based financial institution needed to find a way to design, build, test, and deploy IT solutions faster. By standardizing its cloud approach and adopting Amazon EKS, Nationwide accelerated its development cycles, simplified operations, and reduced its total cost of ownership. Now, it can focus on delivering innovative products and services, driving customer satisfaction and business growth.

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Accelerates

development cycles

Standardizes | Reduced

cloud app



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Facilitates

self-service for developers



Overview

With over 16 million members, Nationwide Building Society (Nationwide) was founded on the belief that its members can achieve more by acting together than they can alone. As the Nationwide community has grown, the importance of digital transformation became apparent. To meet the demands of its members, Nationwide needed to design, build, test, and deploy IT solutions faster.

Turning to Amazon Web Services (AWS), Nationwide built a centralized enterprise platform-as-a-service (ePaaS) solution. This scalable, cloud-native architecture now underpins the development of net-new and existing production workloads, including its contact center, digital customer journeys, open banking confirmation, and much more.



Opportunity | Seeking the Advantages of Cloud Technologies

Headquartered in Swindon, England, Nationwide provides a range of banking and financial services to its members, including loans, credit cards, insurance, and investments. In 2017, the institution's leadership team embarked on a digital transformation journey to strengthen its internal engineering capabilities, adopt new ways of working, simplify its technology footprint, and reduce operational complexity.



As part of this initiative, different teams began to experiment with cloud technologies. As Nationwide observed these efforts, it recognized the advantages of the cloud: teams can quickly deploy new workloads, accelerating development cycles.

“This spurt of innovation and experimentation increased efficiency in pockets, but it lacked a strong enterprise focus,” says Dave Burke, CTO at Nationwide. “Moreover, this also introduced risks and challenges from a controls and governance perspective.”

To standardize these processes, Nationwide began to build ePaaS. It adopted Kubernetes, an open-source container orchestration system, to deploy containers for business workloads. Initially, teams built bespoke Kubernetes-based solutions, but these processes were difficult to scale and manage.

To simplify the ePaaS environment and reduce operational overhead, Nationwide recommissioned the build of secure, centralized, and cloud-native ePaaS models on AWS. It adopted [Amazon Elastic Kubernetes Service](#) (Amazon EKS), the most trusted way to start, run, and scale Kubernetes, to facilitate the federated deployment of its clusters.

Solution | Driving Consistency and Standardization on AWS

With ePaaS, Nationwide standardized its cloud approach while reducing operational overhead. Under the [Shared Responsibility Model](#), AWS is responsible for managing the controls associated with the infrastructure deployed in the AWS environment. Nationwide can configure Amazon EKS as needed to meet its security and governance requirements. It no longer needs to manage container orchestration, update deployment, or workload optimization.

Engineering teams rely on Amazon EKS to handle the heavy lifting of security and operating Kubernetes at scale. The centralized ePaaS integrates with Nationwide’s AWS landing zone, as well as other cloud-native services for monitoring, logging, secrets management, and other capabilities. As a result, developers no longer need to build network and security integrations themselves. And by adopting [Amazon Elastic Compute Cloud](#) (Amazon EC2) [Spot Instances](#), which run fault-tolerant workloads for up to 90 percent off, Nationwide immediately reduced its total cost of ownership.

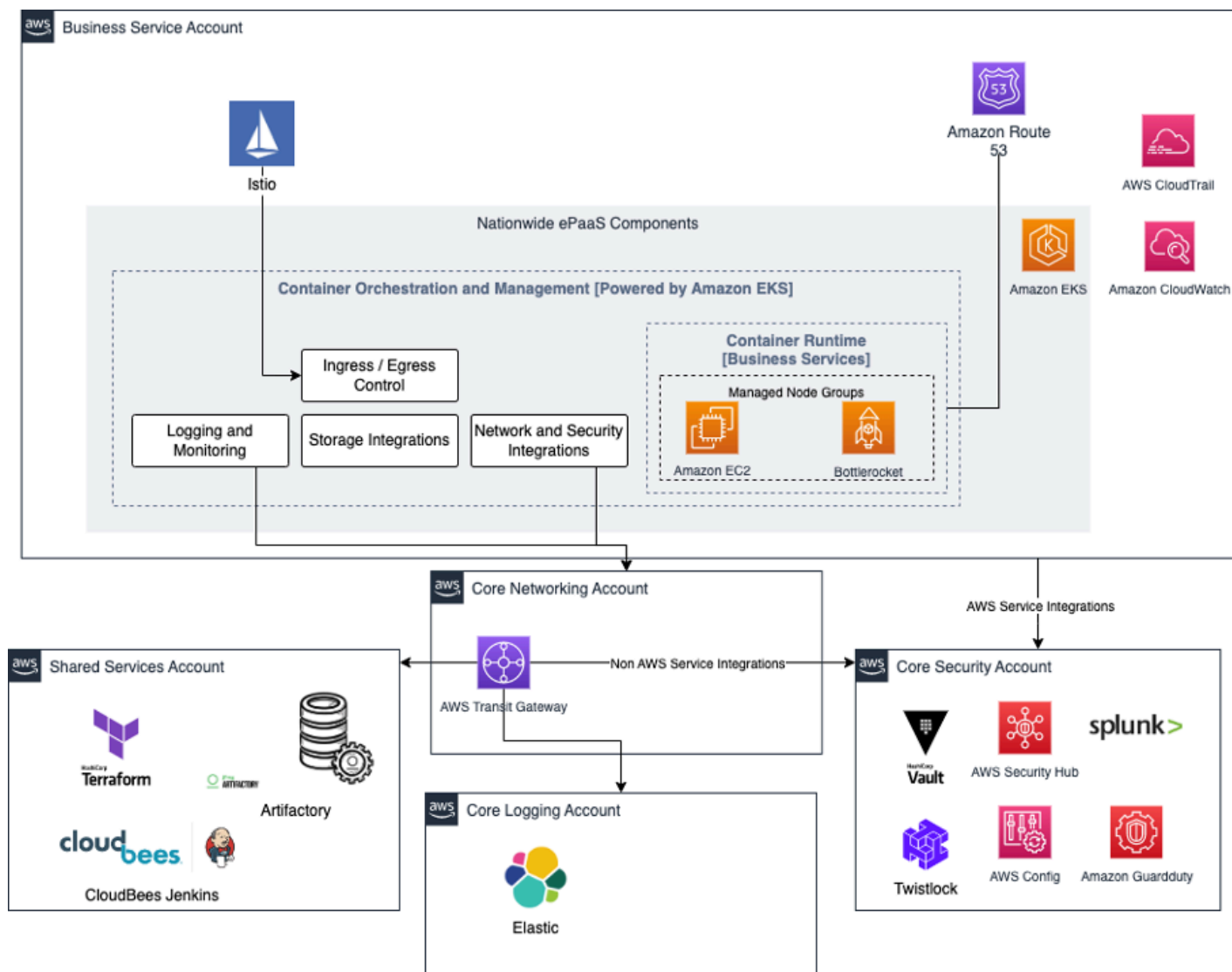
“A standardized ePaaS reduced our duplication of effort. It is deployed in a decentralized manner and maintained centrally,” says Tim Wright, head of cloud operations at Nationwide. “The operational overhead of a self-managed cluster has virtually gone away, which lets us report on compliance against control frameworks across the organization.”



To develop ePaaS, Nationwide implemented best practices, standards, and blueprints created by an internal group. Key principles such as “everything as code” drove further standardization and consistency. Nationwide also adopted [chaos engineering](#), a practice that can help financial services institutions bridge the operational resilience gap. Chaos engineering involves conducting experiments to test the failure resilience of distributed software systems. By doing so, it helps identify system weaknesses. This process is crucial for demonstrating to regulators the capability of production-scale distributed software systems to withstand turbulent phases. With chaos engineering, Nationwide successfully tested and validated resilience for ePaaS-based workloads.

Architecture Diagram

Nationwide ePaaS high-level architecture



Click to enlarge for fullscreen viewing.

Outcome | Becoming Customer Obsessed

To encourage ePaaS adoption, Nationwide established a Cloud Engineering Team and used the Responsibility Model to define the roles and responsibilities of different teams. To ease the onboarding of important initial consumers, it embedded platform engineers in product teams, scaling ePaaS skills across the



organization. Nationwide hosted several [AWS GameDays](#), which are collaborative learning exercises that test skills in implementing AWS solutions, to further improve operational maturity.

Nationwide also implemented several features to facilitate self-service for ePaaS users, including automated patch/feature releases and a postbuild stage where developers can incorporate their own continuous integration/continuous deployment toolchains.

“By moving to a federated self-service model, we helped grow the adoption of cloud technologies within Nationwide,” says Dave Burke. “Within 18 months of being generally available, ePaaS is used by over 15 teams, and there are over 150 Amazon EKS clusters in operation. On AWS, teams can focus less on provisioning and maintaining their infrastructure and more on delivering new, innovative products to members.”

Now, ePaaS hosts critical applications for Nationwide, including mortgage applications, affordability calculators, and credit card originations. The company is actively incorporating inputs from ePaaS consumers to drive the solution’s road map. As it moves forward, this feedback will help ePaaS meet even more of the institution’s business objectives and the needs of its members.

“Amazon EKS has facilitated numerous migrations from legacy cloud systems to ePaaS, providing increased security, resiliency, and scalability to our internal consumers,” says Tim Wright. “In the future, we will expand its hosting potential by supporting more critical business applications.”

About Nationwide Building Society

Nationwide Building Society is a British mutual financial institution headquartered in Swindon, England. With over 16 million members, it is the world’s largest building society with a mission to build a stronger, kinder world for all.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. Access reliable, scalable infrastructure on demand. Scale capacity within minutes with SLA commitment of 99.99% availability.

[Learn more »](#)

Amazon EKS

Amazon EKS is a managed Kubernetes service to run Kubernetes in the AWS cloud and on premises.



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Amazon EC2 Spot Instances

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud. Spot Instances are available at up to a 90% discount compared to On-Demand prices.

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